An Efficient Hamiltonian-within-Gibbs Sampler with Elliptical Slice Sampling and Split–Merge Augmentation for Bayesian Nonparametric Mendelian Randomization

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1 Overview and Motivation

This document proposes a modern, efficient Markov Chain Monte Carlo (MCMC) algorithm for posterior inference in the Bayesian nonparametric Mendelian randomization models BNPM-MR and BNPM-PMR.

The target posterior for BNPM-PMR is

$$p\big(\gamma, \psi, \tau, \beta_{1:p}, \alpha \mid y_{1:p}\big) \;\propto\; \left[\prod_{j=1}^{p} p(y_j \mid \gamma, \psi, \tau, \beta_j)\right] p(\gamma)\, p(\psi)\, p(\tau) \left[\prod_{j=1}^{p} p(\beta_j \mid G)\right] p(G \mid \alpha)\, p(\alpha), \tag{1}$$

where the SNP-specific effects $\beta_1, \ldots, \beta_p$ are induced by a Dirichlet process (DP) prior. Setting $\tau = 0$ yields the BNPM-MR target.

The baseline Gibbs sampler in the paper updates each parameter via (i) an exact Gibbs step for γ, (ii) random-walk Metropolis–Hastings (RWMH) for ψ and τ, (iii) Neal's Algorithm 8 for the DP partition with per-atom RWMH, and (iv) the Escobar–West mixture-of-Gammas update for α. While asymptotically exact, this scheme has three practical inefficiencies:

1. **Inefficient scale-parameter updates**: RWMH proposals for ψ and τ require careful hand-tuning and exhibit diffusive exploration, especially when ψ and τ are a posteriori correlated with the cluster atoms.
2. **Inefficient atom updates**: per-atom RWMH ignores the Gaussian structure of the base measure G_0 and the smoothness of the likelihood in β; proposals must be re-tuned for each dataset.
3. **Slow partition mixing**: sequential, single-site reallocation steps can become trapped in local modes, particularly when two cluster atoms are close in value or when K needs to change.

The proposed algorithm addresses these inefficiencies through three targeted innovations:

1. **No-U-Turn Sampler (NUTS)** [Hoffman and Gelman(2014)] for the scale parameters (ψ, τ): gradient-based HMC proposals that auto-tune step size and trajectory length via dual averaging, entirely replacing manual tuning.
2. **Elliptical Slice Sampling (ESS)** [Murray et al.(2010)] for the cluster atoms β_k^*: a tuning-free, rejection-free sampler that exploits the Gaussian base measure G_0 as a natural auxiliary distribution, guaranteeing acceptance in finitely many shrinkage steps.
3. **Jain–Neal split–merge moves** [Jain and Neal(2004), Jain and Neal(2007)] for the DP partition: non-local transdimensional moves allowing direct transitions between configurations with different numbers of clusters K, dramatically improving mixing across causal mechanism configurations.

The result is a *Hamiltonian-within-Gibbs* sampler augmented with split–merge moves that requires no manual tuning beyond the specification of prior hyperparameters.

2 Notation and Model Summary

Table 1 collects the key notation used throughout.

Table 1: Summary of notation.

Symbol	Description
p	Number of SNPs
$y_j = (\hat{\gamma}_j, \hat{\Gamma}_j)^\top$	Observed summary statistics for SNP j
$(s^2_{\hat{\gamma}_j}, s^2_{\hat{\Gamma}_j})$	Known sampling variances for SNP j
γ	Population mean SNP–exposure effect
$\psi > 0$	Std. deviation of population SNP–exposure effects
$\tau \geq 0$	Systematic pleiotropy std. deviation (BNPM-PMR only)
β_j	SNP-specific causal effect
$G \sim \mathrm{DP}(\alpha, G_0)$	DP mixing measure
$\alpha > 0$	DP concentration parameter
$G_0 = \mathcal{N}(\mu_\beta, \sigma^2_\beta)$	DP base measure
K	Current number of non-empty clusters
$\xi_j \in \{1, \ldots, K\}$	Cluster allocation for SNP j
β^*_k	Atom (causal effect) for cluster k
$n_k = \lvert\{j : \xi_j = k\}\rvert$	Size of cluster k
$\mathcal{C}_k = \{j : \xi_j = k\}$	Index set for cluster k
$\theta = \log\psi,\ \phi = \log\tau$	Log-scale reparameterizations

Marginalized measurement model. After integrating out $\gamma_j \sim \mathcal{N}(\gamma, \psi^2)$, the bivariate marginal distribution of y_j is

$$y_j \mid \gamma, \psi, \tau, \beta_j \ \sim\ \mathcal{N}_2\Big(m_j(\gamma, \beta_j),\ \Sigma_j(\psi, \tau, \beta_j)\Big), \tag{2}$$

with

$$m_j(\gamma, \beta_j) = \begin{pmatrix} 1 \\ \beta_j \end{pmatrix} \gamma, \qquad \Sigma_j(\psi, \tau, \beta_j) = \begin{pmatrix} a_j & c_j(\beta_j) \\ c_j(\beta_j) & d_j(\beta_j, \tau) \end{pmatrix}, \tag{3}$$

where, for a generic atom value b,

$$a_j = \psi^2 + s^2_{\hat{\gamma}_j}, \qquad c_j(b) = b\,\psi^2, \qquad d_j(b, \tau) = b^2\psi^2 + s^2_{\hat{\Gamma}_j} + \tau^2. \tag{4}$$

The determinant and inverse are

$$D_j(b, \tau) := |\Sigma_j| = a_j d_j(b, \tau) - c_j(b)^2, \qquad \Sigma_j^{-1} = \frac{1}{D_j} \begin{pmatrix} d_j & -c_j \\ -c_j & a_j \end{pmatrix}. \tag{5}$$

The per-SNP log-likelihood (constant terms dropped) is

$$\ell_j(b; \gamma, \psi, \tau) = -\tfrac{1}{2} \log D_j(b, \tau) - \tfrac{1}{2}\, r_j(\gamma, b)^\top \Sigma_j^{-1}(b, \tau)\, r_j(\gamma, b), \tag{6}$$

where $r_j(\gamma, b) = (\hat{\gamma}_j - \gamma,\ \hat{\Gamma}_j - b\gamma)^\top$. For BNPM-MR, set $\tau = 0$ everywhere.

3 Gradient Computations

The NUTS update (Section 4.3) requires the partial derivatives of ℓ_j with respect to $\theta = \log\psi$ and (for BNPM-PMR) $\phi = \log\tau$. We also provide the gradient with respect to the atom b for use in optional gradient-based atom updates. All expressions are per-SNP j; indices are suppressed where unambiguous.

Define the shorthand quantities

$$u := \hat{\gamma}_j - \gamma, \qquad v(b) := \hat{\Gamma}_j - b\gamma, \qquad N(b) := d\,u^2 - 2c\,u\,v + a\,v^2, \qquad Q(b) := \frac{N(b)}{D}, \tag{7}$$

so that $r^\top \Sigma^{-1} r = Q$ and $\ell_j = -\frac{1}{2}\log D - \frac{1}{2}Q$.

3.1 Gradient with Respect to the Atom b

The elementary derivatives are $\partial c/\partial b = \psi^2$, $\partial d/\partial b = 2b\psi^2$, $\partial v/\partial b = -\gamma$, giving

$$\frac{\partial D}{\partial b} = 2b\psi^2(a - \psi^2) = 2b\psi^2\, s^2_{\hat{\gamma}_j}, \tag{8}$$

$$\frac{\partial N}{\partial b} = 2b\psi^2 u^2 - 2\psi^2 uv + 2b\psi^2 u\gamma - 2a\gamma v. \tag{9}$$

The full gradient of the log-likelihood is

$$\boxed{\frac{\partial \ell_j}{\partial b} = \frac{1}{2D}\frac{\partial D}{\partial b}(Q-1) - \frac{1}{2D}\frac{\partial N}{\partial b}.} \tag{10}$$

3.2 Gradient with Respect to $\theta = \log\psi$

Using $\partial/\partial\theta = \psi\,\partial/\partial\psi$, with $\partial a/\partial\psi = 2\psi$, $\partial c/\partial\psi = 2b\psi$, $\partial d/\partial\psi = 2b^2\psi$, one obtains

$$\frac{\partial D}{\partial \psi} = 2\psi\big(s^2_{\hat{\Gamma}_j} + b^2 s^2_{\hat{\gamma}_j}\big), \qquad \frac{\partial N}{\partial \psi} = 2\psi\,(bu - v)^2. \tag{11}$$

Multiplying by ψ to convert to the log-scale:

$$\boxed{\frac{\partial \ell_j}{\partial \theta} = \frac{\psi^2\big(s^2_{\hat{\Gamma}_j} + b^2 s^2_{\hat{\gamma}_j}\big)}{D}(Q-1) - \frac{\psi^2\,(bu-v)^2}{D}.} \tag{12}$$

The half-$t(\nu, 0, \sigma)$ prior contributes on the log-scale:

$$\frac{\partial \log p(\theta)}{\partial \theta} = 1 - \frac{(\nu+1)\,e^{2\theta}}{\nu\sigma^2 + e^{2\theta}}. \tag{13}$$

3.3 Gradient with Respect to $\phi = \log\tau$ (BNPM-PMR only)

Since τ enters only through the additive τ^2 in d_j, we have $\partial d/\partial\tau = 2\tau$, $\partial D/\partial\tau = 2\tau a$, $\partial N/\partial\tau = 2\tau u^2$. On the log-scale:

$$\boxed{\frac{\partial \ell_j}{\partial \phi} = \frac{\tau^2}{D}\big(aQ - a - u^2\big).} \tag{14}$$

The prior gradient mirrors (13) with parameters (ν_τ, σ_τ).

4 The Proposed MCMC Algorithm

4.1 Algorithm Overview

One full sweep of the proposed sampler for BNPM-PMR consists of the following steps. Set $\tau = 0$ throughout for BNPM-MR.

Step 1. Exact Gibbs update for γ.

Step 2. NUTS update for $(\theta, \phi) = (\log \psi, \log \tau)$, using the analytic gradients from Section 3.

Step 3. Sequential allocation update for $\xi_{1:p}$ (Neal's Algorithm 8).

Step 4. Elliptical Slice Sampling for atoms β_k^*, $k = 1, \ldots, K$.

Step 5. Split–merge move (every T_{SM} sweeps).

Step 6. Escobar–West update for α (exact).

4.2 Step 1: Exact Gibbs Update for γ

Let $\mathbf{A}_j = (1, b_j)^\top$ with $b_j = \beta^*_{\xi_j}$. With the conjugate prior $\gamma \sim \mathcal{N}(\mu_\gamma, \sigma_\gamma^2)$, the full conditional is

$$\gamma \mid \text{rest} \;\sim\; \mathcal{N}\big(\mu_{\gamma|},\, V_{\gamma|}\big), \tag{15}$$

with

$$V_{\gamma|}^{-1} = \sigma_\gamma^{-2} + \sum_{j=1}^{p} \mathbf{A}_j^\top \Sigma_j^{-1} \mathbf{A}_j, \qquad \mu_{\gamma|} = V_{\gamma|}\left(\sigma_\gamma^{-2}\mu_\gamma + \sum_{j=1}^{p} \mathbf{A}_j^\top \Sigma_j^{-1} y_j\right). \tag{16}$$

This step is exact and costs $\mathcal{O}(p)$.

4.3 Step 2: NUTS Update for (θ, ϕ)

Target log-density.

$$\mathcal{L}(\vartheta) = \sum_{j=1}^{p} \ell_j(b_j; \gamma, e^\theta, e^\phi) + \log p(\theta) + \log p(\phi), \tag{17}$$

where $\vartheta = (\theta, \phi)^\top$ (or just θ for BNPM-MR), and $b_j = \beta^*_{\xi_j}$ is the current atom for SNP j. The gradient $\nabla_\vartheta \mathcal{L}$ is assembled from equations (12), (14), (13), and their counterpart for ϕ.

Leapfrog integrator. Introduce auxiliary momentum $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, M)$ (mass matrix M adapted during warm-up; see Section 6.1). One leapfrog step with step size ε:

$$\tilde{\mathbf{r}} \leftarrow \mathbf{r} + \tfrac{\varepsilon}{2}\, \nabla_\vartheta \mathcal{L}(\vartheta), \tag{18}$$

$$\tilde{\vartheta} \leftarrow \vartheta + \varepsilon\, M^{-1}\tilde{\mathbf{r}}, \tag{19}$$

$$\tilde{\mathbf{r}} \leftarrow \tilde{\mathbf{r}} + \tfrac{\varepsilon}{2}\, \nabla_\vartheta \mathcal{L}(\tilde{\vartheta}). \tag{20}$$

Repeating for L steps (set by NUTS) produces a proposal $(\tilde{\vartheta}, -\tilde{\mathbf{r}})$, accepted with probability $\min(1,\, e^{H_0 - \tilde{H}})$, where $H = -\mathcal{L}(\vartheta) + \frac{1}{2}\mathbf{r}^\top M^{-1}\mathbf{r}$.

No-U-Turn criterion [Hoffman and Gelman(2014)]. The trajectory is terminated when $(\tilde{\vartheta} - \vartheta)^\top \mathbf{r} < 0$ or $(\tilde{\vartheta} - \vartheta)^\top \tilde{\mathbf{r}} < 0$, eliminating the need to choose L manually.

Input: Current (θ, ϕ), atoms $\{b_k\}$, allocations $\{\xi_j\}$, γ; mass matrix M; step size ε

1. Compute $\nabla_{(\theta,\phi)}\mathcal{L}$ from (12)–(14) summed over all SNPs
2. Sample $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, M)$; $\quad H_0 \leftarrow -\mathcal{L}(\theta, \phi) + \frac{1}{2}\mathbf{r}^\top M^{-1}\mathbf{r}$
3. Run NUTS tree-building (leapfrog + No-U-Turn criterion), producing proposal $(\tilde{\theta}, \tilde{\phi})$ with Hamiltonian $\tilde{H}$
4. Set $(\theta, \phi) \leftarrow (\tilde{\theta}, \tilde{\phi})$ with probability $\min(1, e^{H_0 - \tilde{H}})$
5. $\psi \leftarrow e^\theta$; $\tau \leftarrow e^\phi$
6. **(Warm-up only)** Update ε via dual averaging [Nesterov(2009)]; update M via Welford estimator

Algorithm 1. NUTS Update for $(\log \psi, \log \tau)$

Remark 1. For BNPM-MR the update reduces to a one-dimensional NUTS on θ. Even a single-leapfrog HMC step (i.e. MALA) with dual-averaged step size substantially outperforms RWMH for this target.

4.4 Step 3: Sequential Allocation for $\xi_{1:p}$

We retain Neal's Algorithm 8 [Neal(2000)] with m auxiliary atoms. For each SNP $j = 1, \ldots, p$:

1. Remove SNP j from its cluster; delete if empty. Let K^{-j} be the current number of non-empty clusters.
2. Draw m auxiliary atoms $b_1^\circ, \ldots, b_m^\circ \overset{\text{iid}}{\sim} G_0$.
3. Compute unnormalized log-weights

$$\log w_k = \begin{cases} \log n_k^{-j} + \ell_j(b_k; \gamma, \psi, \tau), & k = 1, \ldots, K^{-j}, \\ \log(\alpha/m) + \ell_j(b_r^\circ; \gamma, \psi, \tau), & r = 1, \ldots, m; \end{cases} \tag{21}$$

   normalize using log-sum-exp for numerical stability.
4. Sample ξ_j from the resulting categorical distribution. If a new cluster is chosen, initialize its atom at b_r°.

Full sweep cost: $\mathcal{O}(p(K + m))$ likelihood evaluations.

4.5 Step 4: Elliptical Slice Sampling for Atoms

After the allocation sweep, each atom β_k^* is updated targeting

$$p(\beta_k^* \mid \mathcal{C}_k, \gamma, \psi, \tau) \;\propto\; \mathcal{N}(\beta_k^*; \mu_\beta, \sigma_\beta^2) \prod_{j \in \mathcal{C}_k} p(y_j \mid \gamma, \psi, \tau, \beta_k^*). \tag{22}$$

Why Elliptical Slice Sampling? The base measure $G_0 = \mathcal{N}(\mu_\beta, \sigma_\beta^2)$ is Gaussian, making it the natural auxiliary distribution in the ESS framework [Murray et al.(2010)]. ESS is tuning-free (no step size to set), rejection-free (each call returns an accepted sample), and exploits the Gaussian prior structure via the "ellipse"

$$\beta(\varphi) = (b - \mu_\beta)\cos\varphi + (\nu - \mu_\beta)\sin\varphi + \mu_\beta, \quad \varphi \in [0, 2\pi), \tag{23}$$

where $\nu \sim G_0$ is an auxiliary draw. All points on this ellipse share the same prior density, so acceptance depends only on the likelihood.

Input: Current atom b; cluster $\mathcal{C}_k$; parameters (γ, ψ, τ); $G_0 = \mathcal{N}(\mu_\beta, \sigma_\beta^2)$

1. Draw $\nu \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2)$ ▷ *Gaussian auxiliary point*
2. Draw $u \sim \text{Uniform}(0, 1)$; $\ell^* \leftarrow \sum_{j \in \mathcal{C}_k} \ell_j(b; \gamma, \psi, \tau) + \log u$ ▷ *Likelihood threshold*
3. Draw $\varphi \sim \text{Uniform}(0, 2\pi)$; set bracket $[\varphi_{\min}, \varphi_{\max}] \leftarrow [\varphi - 2\pi,\ \varphi]$
4. **Repeat:**
5. Propose $b^{\text{prop}} \leftarrow (b - \mu_\beta)\cos\varphi + (\nu - \mu_\beta)\sin\varphi + \mu_\beta$ ▷ *see* (23)
6. **If** $\sum_{j \in \mathcal{C}_k} \ell_j(b^{\text{prop}}; \gamma, \psi, \tau) > \ell^*$: $b \leftarrow b^{\text{prop}}$; **return**
7. **Else:** shrink bracket—if $\varphi < 0$ set $\varphi_{\min} \leftarrow \varphi$, else $\varphi_{\max} \leftarrow \varphi$
8. Redraw $\varphi \sim \text{Uniform}(\varphi_{\min}, \varphi_{\max})$; go to Step 5
9. **Until** accepted; $\beta_k^* \leftarrow b$

Algorithm 2. ESS Update for One Cluster Atom β_k^*

Input: Allocations $\{\xi_j\}$; atoms $\{b_k\}_{k=1}^K$; parameters (γ, ψ, τ); G_0

1. **For** $k = 1$ **to** K ▷ *Can be parallelized across clusters*
2. $\mathcal{C}_k \leftarrow \{j : \xi_j = k\}$
3. Run Algorithm 2 with atom b_k and cluster $\mathcal{C}_k$
4. Update $b_k \leftarrow$ new sample from Algorithm 2
5. **End for**

Algorithm 3. ESS Update for All K Cluster Atoms

Remark 2 (Gradient-based alternative). If the atom posterior (22) is suspected to be multimodal (e.g. for very large clusters), one may supplement ESS with a NUTS step on β_k^* using the gradient (10) summed over $j \in \mathcal{C}_k$, together with the Gaussian prior gradient.

4.6 Step 5: Split–Merge Moves

Single-site allocation steps can fail to transition between configurations with different K. Split–merge moves [Jain and Neal(2004), Jain and Neal(2007)] address this by proposing non-local, transdimensional moves.

A split–merge proposal is attempted every T_{SM} full sweeps (recommended: $T_{\text{SM}} = 5$–10).

Setup. Select two distinct SNPs i and j uniformly at random. If $\xi_i = \xi_j = c$: propose a **split** of cluster c. If $\xi_i = c \neq d = \xi_j$: propose a **merge** of c and d.

4.6.1 Split Move

Let $\mathcal{C}_c = \{l : \xi_l = c\}$ with current atom b_c^*.

1. *Initialize launch atoms.* Draw $b_i^{\text{lnch}} \sim G_0$ and $b_j^{\text{lnch}} \sim G_0$ independently. Set $S_i \leftarrow \{i\}$, $S_j \leftarrow \{j\}$.

2. *Launch sequence* (for $t = 1, \ldots, t_{\text{lnch}}$, recommended $t_{\text{lnch}} = 3$–5):

   (a) For each $l \in \mathcal{C}_c \setminus \{i, j\}$ in random order, assign l to S_i or S_j with probabilities proportional to

   $$|S_i| \exp[\ell_l(b_i^{\text{lnch}}; \gamma, \psi, \tau)] \quad \text{and} \quad |S_j| \exp[\ell_l(b_j^{\text{lnch}}; \gamma, \psi, \tau)]. \tag{24}$$

   (b) Update b_i^{lnch} via one ESS step targeting $p(\beta \mid S_i, \gamma, \psi, \tau)$; update b_j^{lnch} similarly.

3. *Record proposal probability.* Denote proposed clusters $\mathcal{C}_{c_1} = S_i$, $\mathcal{C}_{c_2} = S_j$, with atoms $\tilde{b}_{c_1}, \tilde{b}_{c_2}$. Record

   $$\log q_{\text{split}} = \sum_{l \in \mathcal{C}_c \setminus \{i,j\}} \log \tilde{p}_{\text{lnch}}\big(l \to S_{\xi_l^{\text{lnch}}}\big), \tag{25}$$

   where $\tilde{p}_{\text{lnch}}(\cdot)$ is the probability (24) from the final launch sweep.

4. *Accept/reject.* Accept the split with probability $\min(1, A_{\text{split}})$ where

   $$\log A_{\text{split}} = \underbrace{(L_{c_1} + L_{c_2} - L_c)}_{\text{likelihood}} + \underbrace{(P_{c_1} + P_{c_2} - P_c)}_{\text{atom prior}} + \underbrace{\log \alpha + \log \Gamma(n_{c_1}) + \log \Gamma(n_{c_2}) - \log \Gamma(n_c)}_{\text{CRP prior}} - \underbrace{\log q_{\text{split}}}_{\text{proposal}}, \tag{26}$$

   with $L_k = \sum_{l \in \mathcal{C}_k} \ell_l(b_k^*; \gamma, \psi, \tau)$ and $P_k = \log \mathcal{N}(\beta_k^*; \mu_\beta, \sigma_\beta^2)$. If accepted: $K \leftarrow K + 1$, update memberships and atoms.

4.6.2 Merge Move

This is the time-reversal of the split. Define $\mathcal{C}_{\text{new}} = \mathcal{C}_c \cup \mathcal{C}_d$.

1. *Merged atom.* Draw $b_{\text{new}} \sim G_0$, then run one ESS step targeting $p(\beta \mid \mathcal{C}_{\text{new}}, \gamma, \psi, \tau)$.

2. *Compute reverse proposal probability.* Replay the restricted allocation (24) with atoms (b_c^*, b_d^*) for all $l \in \mathcal{C}_{\text{new}} \setminus \{i, j\}$ to obtain $\log q_{\text{split}}^{\text{rev}}$.

3. *Accept/reject.*

   $$\log A_{\text{merge}} = (L_{\text{new}} - L_c - L_d) + (P_{\text{new}} - P_c - P_d) + \log \Gamma(n_c + n_d) - \log \alpha - \log \Gamma(n_c) - \log \Gamma(n_d) + \log q_{\text{split}}^{\text{rev}}. \tag{27}$$

   Note $A_{\text{merge}} = A_{\text{split}}^{-1}$ in the reverse configuration, ensuring detailed balance [Jain and Neal(2004), Jain and Neal(2007)].

4.7 Step 6: Escobar–West Update for α

Under the conjugate prior $\alpha \sim \text{Gamma}(a_\alpha, b_\alpha)$, the full conditional of α given K and p is [Escobar and West(1995)]:

1. Sample $\eta \mid \alpha \sim \text{Beta}(\alpha + 1, p)$.
2. Sample $\alpha \mid K, \eta$ from the mixture

$$g \cdot \text{Gamma}(a_\alpha + K,\ b_\alpha - \log \eta) + (1 - g) \cdot \text{Gamma}(a_\alpha + K - 1,\ b_\alpha - \log \eta), \tag{28}$$

with weight $g/(1 - g) = (a_\alpha + K - 1)/[p(b_\alpha - \log \eta)]$.

Cost: $\mathcal{O}(1)$.

4.8 Full Algorithm

Input: Data $\{y_j, s^2_{\hat{\gamma}_j}, s^2_{\hat{\Gamma}_j}\}_{j=1}^p$; hyperparameters; S_{wu} warm-up iterations; S post-warmup iterations; T_{SM}

1. **Initialize:** $\gamma \leftarrow 0$, $\psi \leftarrow 1$, $\tau \leftarrow 0.01$, $K \leftarrow 1$, $\xi_{1:p} \leftarrow \mathbf{1}$, $b^*_1 \leftarrow 0$, $\alpha \leftarrow 1$
2. **For** $s = 1$ **to** $S_{\text{wu}} + S$:
3. **Step 1:** Sample $\gamma \mid$ rest (exact Gibbs, Sec. 4.2)
4. **Step 2:** Sample $(\log \psi, \log \tau) \mid$ rest (NUTS, Alg. 1)
5. **Step 3:** **For** $j = 1, \ldots, p$: update ξ_j (Neal's Alg. 8 with m auxiliary atoms, Sec. 4.3)
6. **Step 4:** Update b^*_k for $k = 1, \ldots, K$ (ESS, Alg. 3) ▷ *Parallelizable over k*
7. **If** $s \bmod T_{\text{SM}} = 0$: **Step 5:** propose split or merge (Sec. 4.6)
8. **Step 6:** Sample $\alpha \mid K, \text{rest}$ (Escobar–West, Sec. 4.6)
9. **If** $s \leq S_{\text{wu}}$: adapt ε and M (Sec. 6.1)
10. **Else:** store $(\gamma, \psi, \tau, \{\xi_j\}, \{b^*_k\}, K, \alpha)$
11. **End for**

Algorithm 4. Full Proposed MCMC Sampler for BNPM-PMR ($\tau = 0$ for BNPM-MR)

5 Extension to BNPM-PMR

BNPM-PMR augments BNPM-MR with the systematic pleiotropy variance τ^2, which enters only through $d_j(b, \tau) = b^2\psi^2 + s^2_{\hat{\Gamma}_j} + \tau^2$. The proposed algorithm requires only two modifications:

1. **NUTS operates on $(\theta, \phi) \in \mathbb{R}^2$ jointly.** The mass matrix M adapts during warm-up to capture any posterior correlation between ψ and τ, automatically improving proposal geometry.
2. **All likelihood evaluations** use $d_j(b, \tau)$; gradients (12) and (14) are used as written.

A practical identifiability concern: both τ^2 and within-cluster dispersion of $\{\beta^*_k\}$ can explain variability in $\hat{\Gamma}_j$. The joint NUTS update for (ψ, τ) and the DP prior on α provide a coherent solution: any ridge in the (ψ, τ) marginal posterior is explored along its principal axis by the adapted mass matrix, while the cluster structure is simultaneously updated in Steps 3–5.

6 Implementation Details

6.1 Warm-Up and Adaptation

Step size adaptation. Use Nesterov's dual-averaging scheme [Nesterov(2009)] to adapt ε toward mean acceptance probability $\delta^* = 0.65$:

$$\log \bar{\varepsilon}_s = \mu - \frac{\sqrt{s}}{t_0} \sum_{t=1}^{s} \frac{\delta^* - a_t}{s\, t^{\kappa}}, \tag{29}$$

where a_t is the realized HMC acceptance probability, and typical defaults are $\mu = \log(10\varepsilon_0)$, $t_0 = 10$, $\kappa = 0.75$.

Mass matrix. After $S_{\text{wu}}/2$ burn-in iterations, estimate M via the online Welford estimator applied to (θ, ϕ) samples. A dense 2×2 matrix is feasible and recommended when ψ and τ are correlated.

Warm-up schedule. Recommended: $S_{\text{wu}} = 1000$ total; first 500 iterations adapt ε only; second 500 also estimate M.

Auxiliary atoms. Default $m = 3$ for Neal's Algorithm 8; $m = 1$ suffices when K is large.

6.2 Computational Cost

Table 2: Per-sweep computational cost of each step.

Step	Cost	Notes
Gibbs for γ	$\mathcal{O}(p)$	Exact; matrix-vector products
NUTS for (ψ, τ)	$\mathcal{O}(pL)$	$L \leq 10$ typical; gradient each step
Allocation sweep	$\mathcal{O}(p(K+m))$	Sequential; inner loop vectorizable
ESS for atoms	$\mathcal{O}(K n_{\max} I_{\max})$	$n_{\max}$: largest cluster; $I_{\max} \lesssim 10$; parallelizable over k
Split–merge	$\mathcal{O}(n_c\, t_{\text{lnch}})$	n_c: size of affected cluster
Escobar–West	$\mathcal{O}(1)$	Exact

For typical MR datasets ($p = 60$–200, $K \leq 6$), the full sampler runs at > 1000 iterations per second on a single core in a compiled language (C++/Julia).

6.3 Parallelization

1. **Atom updates (Step 4)**: atoms $\{\beta_k^*\}_{k=1}^K$ are conditionally independent given $(\gamma, \psi, \tau, \xi_{1:p})$; all K ESS transitions can be dispatched to separate threads.
2. **Likelihood evaluations**: the p terms in the NUTS gradient and the $p(K+m)$ terms in the allocation sweep are independent across SNPs and can be vectorized.

6.4 Numerical Stability

- **Log-sum-exp** for categorical weights in Step 3.
- **Determinant**: assert $D_j > \epsilon$ (e.g. $\epsilon = 10^{-12}$).
- **ESS**: cap the shrinkage loop at 100 iterations.
- **NUTS near** $\tau = 0$: clamp $\tau \geq 10^{-6}$ during warm-up.

6.5 Convergence Diagnostics

Continuous parameters. Run ≥ 4 parallel chains from overdispersed starting values. Monitor $\hat{R} < 1.01$ and bulk ESS > 400 for $(\gamma, \psi, \tau, \alpha)$ [Vehtari et al.(2021)]. Additionally:

- NUTS divergence rate (target $< 0.5\%$).
- E-BFMI > 0.2 (values below indicate poor posterior geometry).

Partition. Monitor $K^{(s)}$ across iterations; a well-mixing chain should show multiple transitions between different K values.

- *PSM convergence*: compare PSMs from two chains via $\|\widehat{\mathrm{PSM}}_1 - \widehat{\mathrm{PSM}}_2\|_F/p < 0.05$.
- *Split–merge acceptance*: target 20–40% for both moves; rates below 5% suggest t_{lnch} is too small.

6.6 Post-Processing: Label Switching

Cluster labels are non-identifiable across MCMC iterations. Use label-invariant summaries:

- **PSM**: $\widehat{\mathrm{PSM}}_{ij} = S^{-1} \sum_s \mathbf{1}\{\xi_i^{(s)} = \xi_j^{(s)}\}$.
- **Distribution of** K: $\hat{p}(K = k) = S^{-1} \sum_s \mathbf{1}\{K^{(s)} = k\}$.
- **BMA effect**: $\hat{\beta}_j = S^{-1} \sum_s b^{*(s)}_{\xi_j^{(s)}}$.
- **Point partition**: Dahl [Dahl(2006)] or SALSO [Dahl et al.(2022)].

7 Summary of Improvements

The three principal gains are:

1. Elimination of all manual tuning via NUTS dual averaging (scale parameters) and parameter-free ESS (atoms).
2. Better mixing for (ψ, τ) via gradient-informed proposals that adapt to posterior geometry and correlations.
3. Faster exploration of the partition space through split–merge moves, enabling the chain to discover the correct number of causal mechanisms K from any initialization.

Table 3: Baseline vs. proposed sampler.

Component	Baseline	Proposed
γ	Exact Gibbs	Exact Gibbs (unchanged)
ψ	RWMH on $\log\psi$; manual tuning	NUTS; auto-tuned
τ	RWMH on $\log\tau$; manual tuning	Joint NUTS on $(\log\psi, \log\tau)$; mass matrix adapts
$\xi_{1:p}$	Neal's Alg. 8 only	Neal's Alg. 8 + split–merge
β_k^*	Per-atom RWMH; hand-tuned	ESS; tuning-free
α	Escobar–West (exact)	Escobar–West (unchanged)
Partition moves	Local single-site	Local + global split–merge
Tuning	Manual throughout	Fully automated

A Explicit Gradient Formulas for Reference Implementation

Set $a = \psi^2 + s_{\hat{\gamma}}^2$, $c = b\psi^2$, $d = b^2\psi^2 + s_{\hat{\Gamma}}^2 + \tau^2$, $D = ad - c^2$, $u = \hat{\gamma} - \gamma$, $v = \hat{\Gamma} - b\gamma$, $Q = (du^2 - 2cuv + av^2)/D$.

$$\ell = -\tfrac{1}{2}\log D - \tfrac{1}{2}(du^2 - 2cuv + av^2)/D, \tag{30}$$

$$\frac{\partial \ell}{\partial b} = \frac{b\psi^2 s_{\hat{\gamma}}^2}{D}(Q-1) - \frac{b\psi^2 u^2 - \psi^2 uv + b\psi^2 u\gamma - av\gamma}{D}, \tag{31}$$

$$\frac{\partial \ell}{\partial \theta} = \frac{\psi^2(s_{\hat{\Gamma}}^2 + b^2 s_{\hat{\gamma}}^2)}{D}(Q-1) - \frac{\psi^2(bu - v)^2}{D}, \tag{32}$$

$$\frac{\partial \ell}{\partial \phi} = \frac{\tau^2}{D}(aQ - a - u^2). \tag{33}$$

Sum over all $j = 1, \ldots, p$ and add prior gradients (13) to obtain the full NUTS gradient.

B Detailed Balance for Split–Merge Moves

Let the joint density of the augmented state (partition + atoms) be

$$\pi(\rho) \;\propto\; \alpha^K \prod_{k=1}^{K}\Big[\Gamma(n_k)\, p(\beta_k^*) \prod_{j\in\mathcal{C}_k} p(y_j \mid \gamma, \psi, \tau, \beta_k^*)\Big]. \tag{34}$$

The MH ratio for the split equals A_{split} (26) by direct computation of $\pi(\rho')/\pi(\rho) \times q(\rho' \to \rho)/q(\rho \to \rho')$. The merge ratio $A_{\text{merge}} = A_{\text{split}}^{-1}$ follows by reversibility [Jain and Neal(2004)]. The non-conjugate extension uses the atom-marginalized acceptance criterion of [Jain and Neal(2007)].

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